ΠΑΝΕΠΙΣΤΗΜΙΟ ΔΥΤΙΚΗΣ ΑΤΤΙΚΗΣ

UNIVERSITY OF WEST ATTICA

SIGNALS AND SYSTEMS

TASK 1

NAME: ATHANASIOU VASILEIOS EVANGELOS

STUDENT ID: 19390005

SEMESTER: D

LABORATORY DEPARTMENT: DEPARTMENT 1 FRIDAY 9-11

EXERCISE 1

Commands :

```
a = [0:0.1:10];
b = cos ( 0:0.2:20 );
c = a/b
d = a .^ 4
e = a*b'
```

Results :

```
c =

0.8941


d =

1.0e+04 *

Columns 1 through 6

0 0.0000 0.0000    0.0000    0.0000    0.0000

Columns 7 through 12

0.0000 0.0000    0.0000 0.0001 0.0001    0.0001

Columns 13 through 18

0.0002 0.0003 0.0004 0.0005 0.0007 0.0008

Columns 19 through 24

0.0010 0.0013 0.0016 0.0019 0.0023 0.0028

Columns 25 through 30

0.0033 0.0039 0.0046 0.0053 0.0061 0.0071

Columns 31 through 36

0.0081 0.0092 0.0105 0.0119 0.0134 0.0150

Columns 37 through 42

0.0168 0.0187 0.0209 0.0231 0.0256 0.0283

Columns 43 through 48

0.0311 0.0342 0.0375 0.0410 0.0448 0.0488
```

```
Columns 49 through 54

0.0531 0.0576 0.0625 0.0677 0.0731 0.0789

Columns 55 through 60

0.0850 0.0915 0.0983 0.1056 0.1132 0.1212

Columns 61 through 66

0.1296 0.1385 0.1478 0.1575 0.1678 0.1785

Columns 67 through 72

0.1897 0.2015 0.2138 0.2267 0.2401 0.2541

Columns 73 through 78

0.2687 0.2840 0.2999 0.3164 0.3336 0.3515

Columns 79 through 84

0.3702 0.3895 0.4096 0.4305 0.4521 0.4746

Columns 85 through 90

0.4979 0.5220 0.5470 0.5729 0.5997 0.6274

Columns 91 through 96

    0.6561 0.6857 0.7164 0.7481 0.7807 0.8145

  Columns 97 through 101

0.8493 0.8853 0.9224 0.9606 1.0000


e =

46.0507
```

Documentation:

The exercise asks us to create vector " a " with interval from 0-10 and step 0.1, as well as vector " b " with interval plus(0)-plus(20) and step plus(0.2). The request is implemented with the following commands:

```
a = [0:0.1:10]?
b = cos ( 0:0.2:20 );
```

It also asks us to calculate the division of the two vectors (a / b), the square of the vector " a " (a ^4) and the inner product of the two vectors. The results are recorded in the variables “ c ”, “ d ” and “ e ” respectively. The request is implemented with the following commands:

```
c = a / b
d = a.^4
e = a*b'
```

To calculate the inner product we used the inverse line vector " b " as the second term, so that the result is a pure number.

EXERCISE 2

Commands :

```
function [ moires ] = gwnia ( rad )
moires = rad *180/pi
```

Results :

For p/4:

```
>> function( pi/4)

moires =

45


ans =

45

>>
```

Documentation:

The exercise asks us to write a function that will take as an argument a number in radians (rad) and return its value in degrees. The request is implemented with the following commands:

```
function [ moires ] = gwnia ( rad )
moires = rad *180/ pi
```

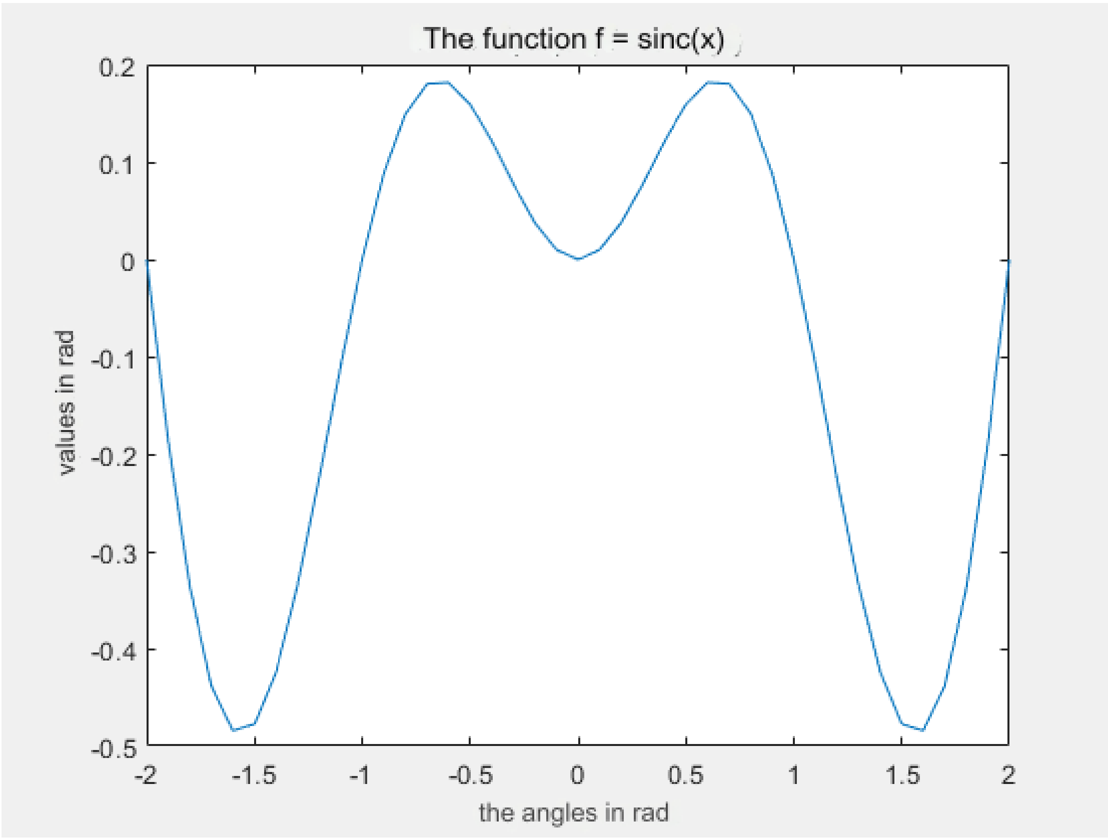
Using the function we find that π/4 rad is 45 degrees.

EXERCISE 3

Commands :

```
t = -2:0.1:2;
f = sin( pi.*t)/pi.*t;
plot ( t,f )
title ( ' H function f = sinc (x)' )
xlabel ( ' the corners in rad ' )
ylabel ( 'values in rad' )
```

Graph:



Documentation:

The exercise asks us to write a function that plots the function sinc (x) = sin (π x) / π x for the interval -2π to 2π. The request is implemented with the following commands:

```
t = -2:0.1:2;
f = sin ( pi .* t )/ pi .* t ;
```

```
plot ( t , f )
```

The ' plot ' function was used to plot the sinc (x) function.

The commands:

```
title ( 'The function f = sinc ( x )' )
xlabel ( 'the angles in rad ' )
ylabel ( 'values in rad' )
```

were used to add labels to the horizontal and vertical axis of the plot (xlabel and ylabel commands) as well as a title to the entire plot (title command).

EXERCISE 4

Commands :

```
function [ a,b,c,d ] = praxis (z)
a = angle( z)
b = abs( z)
c = real( z)
d = imag ( z )
```

Results:

For i :

```
>> function( i )

a =

1.5708


b =

1


c =

0


d =

1
```

```
ans =
1.5708
>>
```

<u>For - i :</u>

```
>> function(- i )
a =
-1.5708

b =
1

c =
0

d =
-1

ans =
-1.5708
>>
```

<u>For 1:</u>

```
>> function( 1 )
a =
0

b =
1

c =
1
```

```
d =

0

ans =

0

>>
```

For e^3+4i:

```
>> function( exp(3+4i))

a =

0.1966

b =

20.4799

c =

20.0855

d =

4

ans =

0.1966

>>
```

Documentation:

The exercise asks us to write a function that will take a complex number as an argument and return the phase, magnitude, real and imaginary part of the complex. The request is implemented with the following commands:

```
function [ a , b , c , d ] = praxis ( z )
a = angle( z)
b = abs( z)
c = real ( z )
d = imag ( z ) In detail, the "
```

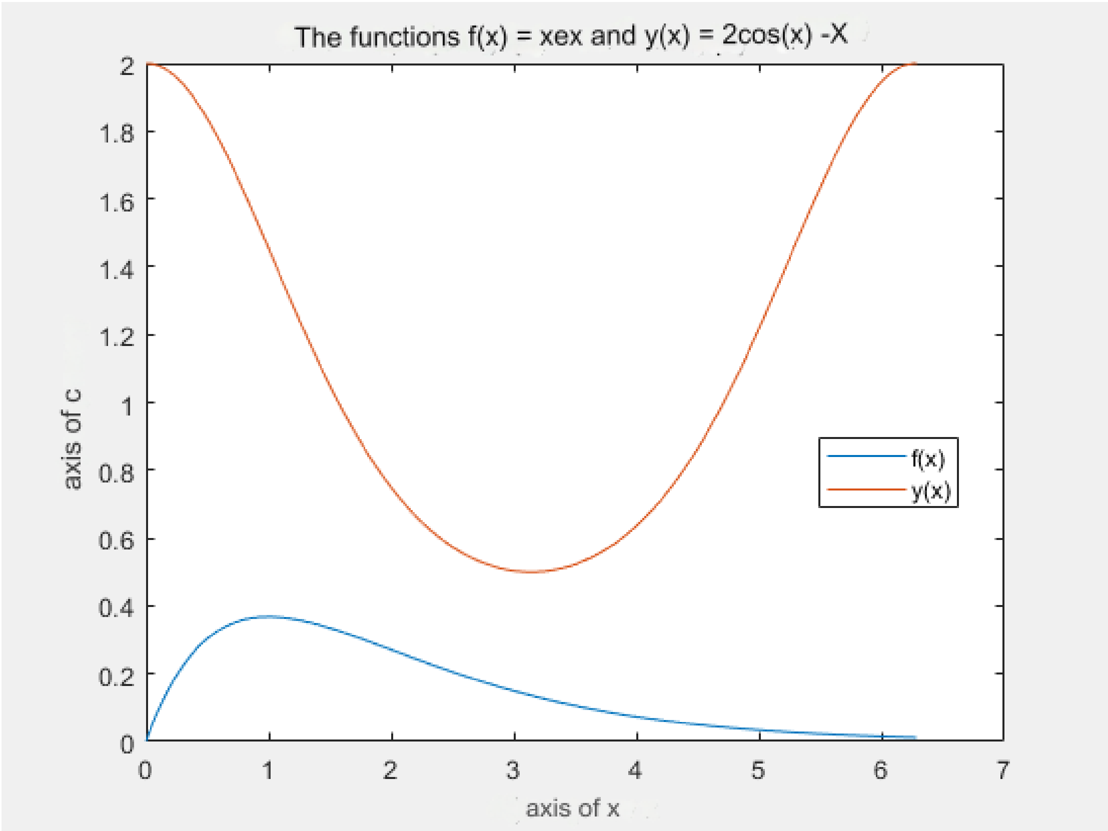
angle " function calculates the phase, " abs " the measure, " real " the real part and " imag " the imaginary part of the complex. The results are recorded in the variables “ a ”, “ b ”, “ c ” and “ d ” respectively.

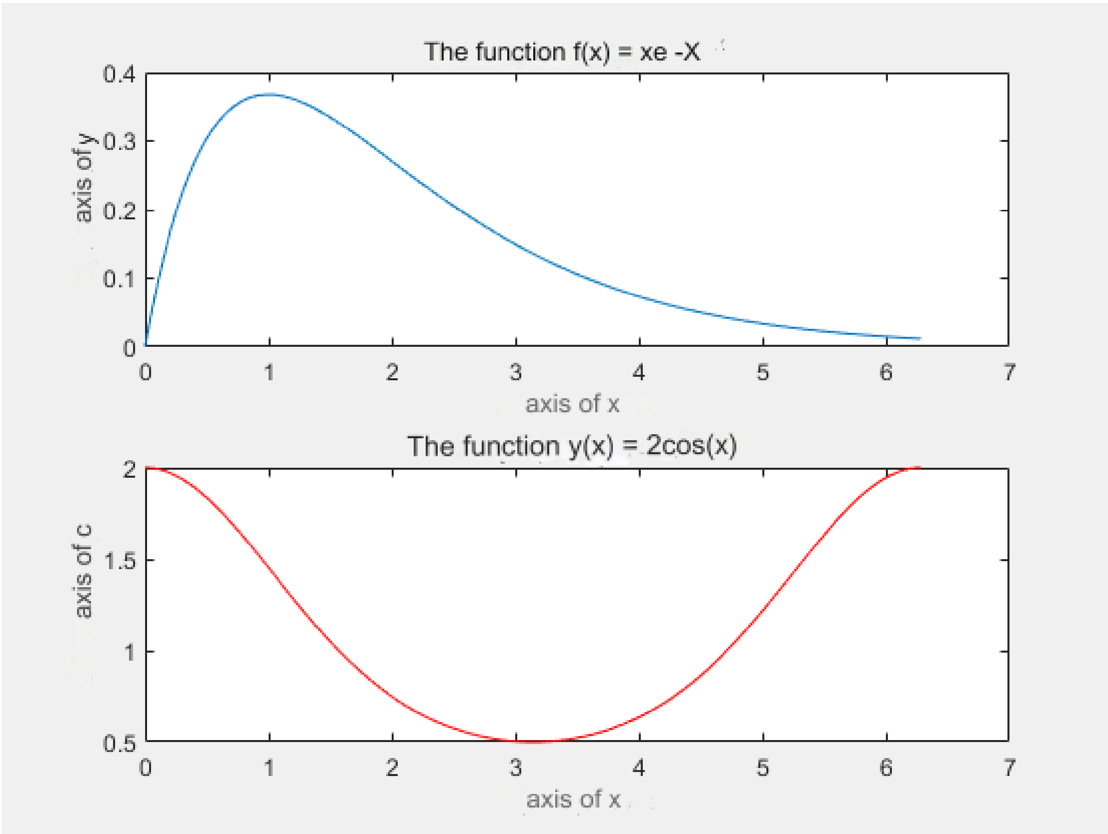
EXERCISE 5

Commands :

```
x = linspace (0.2*pi,500);
f = x.* exp( -x);
y = 2 .^ cos(x);
figures (1)
plot ( x,f,x,y )
title ( 'O i functions f(x) = xe ^-^x and y(x) =
2^c^o^s^(^x^)' )
xlabel ( ' axis of x' )
ylabel ( ' axis of y' )
legend ( 'f(x)' , 'y(x)' )
figures (2)
subplot (2,1,1)
plot ( x,f )
title ( 'The function f(x) = xe^-^x' )
xlabel ( 'x-axis' )
ylabel ( 'y-axis' )
subplot (2,1,2)
plot ( x,y, 'r ' )
title ( 'The function y(x) = 2^c^o^s^(^x^)' )
xlabel ( 'x-axis' )
ylabel ( 'y-axis' )
```

Graphical representations:


The functions f(x) = xex and y(x) = 2cos(x) -X
axis of c
axis of x
f(x)
y(x)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0
1
2
3
4
5
6
7


The function f(x) = xe -X
axis of y
axis of x
0
0.1
0.2
0.3
0.4
0
1
2
3
4
5
6
7
The function y(x) = 2cos(x)
axis of c
axis of x
0.5
1
1.5
2
0
1
2
3
4
5
6
7

Documentation:

The exercise asks us to divide the interval [0 2π] into 500 points. The request is implemented using the " linspace " function:

```
x = linspace (0.2* pi .500);
```

It also requests that we draw in this space (in the same figure) the following signals:

$$f(x) = xe^{-x}, 0 < x < 2\pi$$
$$y(x) = 2^{\cos(x)}, 0 < x < 2\pi$$

where the request is implemented with the following commands:

```
f = x .* exp ( - x );
y = 2 .^ cos(x);
figures (1)
plot ( x,f,x,y )
```

For the title on the plot, the labels on the x- and y- axis, and the label for all the curves, the following commands were used:

```
title ( ' O ι functions f ( x ) = xe ^-^ x and y ( x ) = 2 ^ c
^ o ^ s ^(^ x ^)' )
xlabel ( ' x -axis ' )
ylabel ( ' y -axis ' )
legend ( 'f(x)' , 'y(x)' )
```

Finally, the exercise asks us to redraw the above two signals in a second figure but not 2 different windows. The request is implemented with the " subplot " function, where we divide the active graphics window into 2 x 1 positions, and the following commands:

```
figure (2)
subplot (2,1,1)
plot ( x,f )
title ( ' H function f(x) = xe ^-^x' )
xlabel ( 'x-axis' )
ylabel ( 'y-axis' )
subplot (2,1,2)
plot ( x,y, 'r ' )
title ( 'The function y(x) = 2^c^o^s^(^x^)' )
xlabel ( 'x-axis' )
ylabel ( 'y-axis' )
```

In the second function " plot " we defined the color of the graph of the function " y " with the symbol " r ".

EXERCISE 6

Commands :

```
syms x t ;
f = cos (x) + sin( t) + exp(t);
dx = diff ( f,x )
dt = diff ( f,t )
sym t ;
g = t * exp( -t);
int (g,t,0,inf)
syms x t ;
dg = x^3 * exp(t);
int ( dg,x );
int ( x,t )
syms x k ?
a = x^k / factorial (k);
symsum (a,k,0,inf)
sym x ;
b = 3 / (x+2) + x / (x^2 + 1);
[ arithmitis paranomastis ] = numden (b)
sym x ;
solve (x^3 + 2*x^2 - x - 2)
```

Results:

a)

```
dx =

-sin(x)


dt =

cos ( t) + exp(t)
```

b)

```
ans =

1
```

c)

```
ans =
```

```
t* x
```

d)

```
ans =

exp( x)
```

e)

```
arithmetic =

4*x^2 + 2*x + 3


paranomastis =

( x^ 2 + 1)*(x + 2)
```

f)

```
ans =

-2
-1
 1
```

<u>Documentation:</u>

The exercise in the first subquestion asks us to calculate the partial derivatives of the function. $f(x,t) = \sigma\upsilon\nu(x) + \eta\mu(t) + e^t$ The request is implemented with the following commands:

```
syms x t ;
f = cos (x) + sin( t) + exp(t);
dx = diff ( f,x )
dt = diff ( f,t )
```

In the second sub-question it asks us to calculate the integral $\int_0^\infty te^{-t}dt$. The request is implemented with the following commands:

```
sym t ;
g = t * exp ( - t );
int ( g , t ,0, inf )
```

In the third sub-question it asks us to calculate the double indefinite integral $\iint x^3 e^t dxdt$. The request is implemented with the following commands:

```
syms x t ;
dg = x^3 * exp(t);
int ( dg,x );
int ( x , t )
```

In the fourth sub-question, he asks us to calculate the sum $\sum_{k=0}^{\infty} \frac{x^k}{k!}$. The request is implemented with the following commands:

```
syms x k ?
a = x^k / factorial (k);
symsum ( a , k ,0, inf )
```

The fifth sub-question asks to convert the representation into an explicit one $f(x) = \frac{3}{x+2} + \frac{x}{x^2+1}$. The request is implemented with the following commands:

```
sym x ;
b = 3 / ( x +2) + x / ( x ^2 + 1);
[ arithmitis paranomastis ] = numden ( b )
```

The sixth and final sub-question asks to solve the cubic equation $f(x) = x^3 + 2x^2 - x - 2$. The request is implemented with the following commands:

```
sym x ;
solve ( x ^3 + 2* x ^2 - x - 2)
```